Presentation
UserInterface
+start()
+displayError(message: String)
GUIApplication
CLIApplication
ProfileView
ResearchView
NewsView
ReaderView
LibraryView
WorkspaceView
Controllers
ProfileController
+getReaderProfile()
+updateReaderProfile()
+correctKnowledgeAssessment()
ResearchController
+research(request)
+askQuestion(question)
+continueResearch(sessionId, request)
NewsController
+getDashboard()
+openEvent(eventId)
+researchEvent(eventId)
ReadingController
+openDocument(documentId)
+getAnnotations(passageId)
+generateAnnotation(passageId)
LibraryController
+search(query)
+browse(filter)
+download(documentId)
WorkspaceController
+save(item)
+remove(itemId)
+getWorkspace()
publishes
publishes
Reading and Learning
AnnotationService
+generateAnnotation(passage, type, profile)
+getAnnotations(passageId)
IdeologicalContextService
+findRelevantTraditions(topic)
+explainHistoricalContext(topic, tradition)
+recommendRepresentativeTexts(context)
PrerequisiteService
+identifyPrerequisites(topic)
+buildLearningPath(topic, profile)
RecommendationService
+recommendReadings(topic, profile)
+recommendNext(profile)
LibraryService
+search(query)
+browse(filter)
+getDocument(id)
+download(id)
News System
NewsService
+getDashboardEvents()
+getEvent(id)
+refreshNews()
Event
+id: UUID
+title: String
+summary: String
+lastUpdatedAt: DateTime
EventClusterer
+cluster(articles)
+isSameEvent(a, b)
NewsSummaryService
+summarize(event)
Reader Model
«Subject»
LearningEventBus
+subscribe(observer)
+unsubscribe(observer)
+publish(event)
0..*
«Observer»
ActivityObserver
+onActivity(event)
ReaderProfile
+id: UUID
+preferredDepth: ExplanationDepth
KnowledgeProfile
+getLevel(topic)
+updateLevel(topic, level)
+identifyBlindSpots()
InterestProfile
+addInterest(interest)
+getTopInterests()
ReadingHistory
+addRecord(record)
+getCompletedDocuments()
0..*
PersonalizationService
+personalizePrompt(prompt, profile)
+determineExplanationDepth(profile)
+filterKnownPrerequisites(items, profile)
«Concrete Observer»
ReaderModelUpdater
+onActivity(event)
KnowledgeArea
+topic: String
+level: KnowledgeLevel
+confidence: float
ReaderProfileService
+getProfile(userId)
+updateProfile(profile)
+recordActivity(event)
Workspace
WorkspaceService
+getWorkspace(userId)
+saveItem(userId, item)
+removeItem(userId, itemId)
Workspace
+id: UUID
+userId: UUID
+add(item)
+remove(itemId)
+search(query)
0..*
SavedItem
+id: UUID
+type: SavedItemType
+referenceId: UUID
+title: String
+notes: String
Persistence
ResearchRepository
+findSession(id)
+saveSession(session)
+saveResult(result)
ReaderProfileRepository
+findByUserId(userId)
+save(profile)
DocumentRepository
+findById(id)
+search(query)
+save(document)
WorkspaceRepository
+findByUserId(userId)
+save(workspace)
SQLResearchRepository
SQLReaderProfileRepository
SQLDocumentRepository
SQLWorkspaceRepository
AI Agent
«Facade»
PoliticalResearchAgent
+research(request): ResearchResult
+answerQuestion(question): AgentResponse
+continueResearch(sessionId, request): ResearchResult
-initializeSession(request): ResearchSession
-executeNextStep(session)
-updateResearchState(session)
-shouldContinue(session): boolean
-finalizeResearch(session): ResearchResult
revises plan
ResearchPlanner
+createInitialPlan(request, profile): ResearchPlan
+revisePlan(plan, assessment): ResearchPlan
+identifyNextQuestions(situation): List<String>
DialecticalMaterialistResearchMethod
+assess(situation): ResearchAssessment
+identifyMissingRelations(situation)
+identifyContradictions(situation)
produces
+identifyHistoricalQuestions(situation)
+identifyConflictingEvidence(situation)
+evaluateDevelopment(situation)
+synthesize(situation): MaterialistAnalysis
ConcreteSituationBuilder
+buildInitialSituation(corpus): ConcreteSituation
+updateSituation(situation, evidence)
+integrateRelation(situation, relation)
+integrateHistoricalDevelopment(situation, development)
+integrateContradiction(situation, contradiction)
PromptBuilder
+buildResearchPrompt(context)
+buildQuestionPrompt(context)
+buildAnnotationPrompt(context)
LLMClient
+generate(prompt): LLMResponse
creates / revises
ContradictionAnalyzer
+identifyContradictions(situation): List<Contradiction>
+investigateContradiction(contradiction, situation)
+evaluateDevelopment(contradiction)
EvidenceEvaluator
+evaluateEvidence(evidence)
+findContradictoryEvidence(situation)
+testExplanation(hypothesis, evidence)
+assessConfidence(claim)
MaterialRelationAnalyzer
+identifyRelations(situation): List<MaterialRelation>
+analyzeClassRelations(situation)
+analyzeProductionRelations(situation)
+analyzeInstitutionalRelations(situation)
+analyzeIdeologicalRelations(situation)
HistoricalDevelopmentAnalyzer
+traceOrigins(situation)
+identifyStages(situation)
+identifyContinuities(situation)
+identifyTurningPoints(situation)
+identifyStructuralChanges(situation)
«Adapter»
OpenAILLMAdapter
+generate(prompt): LLMResponse
assessed by
Research Execution
ResearchStoppingPolicy
+shouldStop(session, assessment, budget): boolean
+budgetExceeded(session, budget): boolean
+sufficientCoverage(assessment): boolean
+noUsefulNextSteps(assessment): boolean
checked by
ResearchAssessment
+missingRelations
+historicalQuestions
+contradictions
+conflictingEvidence
+unresolvedQuestions
+requiresMoreResearch: boolean
ResearchSession
+id: UUID
+status: ResearchStatus
+iterationCount: int
+toolCallCount: int
+startedAt: DateTime
+incrementIteration()
+incrementToolCalls()
+complete()
+fail()
creates / revises
ResearchBudget
+maxIterations: int
+maxToolCalls: int
+maxDurationSeconds: int
+maxLLMCalls: int
ResearchResult
+summary: String
+createdAt: DateTime
+getCitations()
+getRecommendedReadings()
ResearchPlan
+objective: String
+addCommand(command)
+getNextCommand()
+replaceCommands(commands)
+isComplete()
schedules
1..*
MaterialistAnalysis
+materialConditions: String
+historicalDevelopment: String
+classRelations: String
+productionRelations: String
+contradictions: String
+institutionalContext: String
+ideologicalContext: String
+unresolvedQuestions: String
«Command»
ResearchCommand
+execute(context)
+retry()
executes
RetrieveTheoryCommand
CompareSourcesCommand
RetrieveEvidenceCommand
InvestigateRelationCommand
TraceHistoricalDevelopmentCommand
InvestigateContradictionCommand
TestExplanationCommand
ResearchContext
+request
+readerProfile
+situation
+corpus
SynthesizeResearchCommand
ConcreteSituation
+topic: String
+materialConditions
+actors
+institutions
+productiveForces
+productionRelations
+historicalProcesses
+ideologicalFactors
+unresolvedQuestions
+addRelation(relation)
+addContradiction(contradiction)
+addEvidence(evidence)
+addHistoricalDevelopment(development)
0..*
0..*
0..*
HistoricalDevelopment
+origins: String
+stages: String
+continuities: String
+turningPoints: String
+structuralChanges: String
+addEvidence(evidence)
MaterialRelation
+sourceEntity: String
+relationType: RelationType
+targetEntity: String
+historicalPeriod: String
+confidence: float
+addEvidence(evidence)
Contradiction
+poleA: String
+poleB: String
+relationship: String
+historicalOrigin: String
+currentForm: String
+development: String
+significance: String
+confidence: float
+addEvidence(evidence)
ResearchCorpus
+topic: String
+addDocument(document)
+addPassage(passage)
+addCitation(citation)
+addEvidence(evidence)
0..*
1..*
0..*
1..*
1..*
Evidence
+id: UUID
+claim: String
+type: EvidenceType
+confidence: float
+interpretationStatus: InterpretationStatus
+addCitation(citation)
1..*
Citation
Citation
0..*
Documents and Sources
Citation
+id: UUID
+locator: String
+quotedText: String
SourceComparison
+sharedFacts: String
+framingDifferences: String
+omittedContext: String
+terminologyDifferences: String
+materialDifferences: String
+compare(documents)
2..*
Document
0..*
+id: UUID
+title: String
+author: String
+publicationDate: Date
+sourceUrl: String
+license: String
+downloadable: boolean
0..1
1..*
0..*
0..1
Book
Passage
+id: UUID
+text: String
Article
SourceProfile
+ownership: String
+fundingModel: String
+institutionalAffiliation: String
+intendedAudience: String
+historicalContext: String
0..*
Annotation
+type: AnnotationType
+content: String
NewsArticle
1..*
+outlet: String
Tools and External Services
ToolManager
+executeTool(name, input): ToolResult
+registerTool(tool)
+getTool(name): Tool
1..*
Tool
+validate(input): boolean
+execute(input): ToolResult
«Factory Method»
ToolFactory
+createTool(type): Tool
WebSearchTool
LibrarySearchTool
HistoricalSourceTool
TheorySearchTool
NewsSearchTool
WebProvider
+search(query)
DocumentProvider
+search(query)
+getDocument(id)
+download(id)
NewsProvider
+searchNews(query)
+getRecentNews()
«Adapter»
WebSearchAPIAdapter
«Adapter»
PublicTextArchiveAdapter
«Adapter»
ExternalNewsAPIAdapter